Black Holes

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Introduction to Black Holes

A black hole is a region of space where gravity is so strong that nothing, not even light, can escape from it. Because light cannot get out, the object cannot be seen directly and appears black, which is why it is called a black hole. The idea follows from Albert Einstein's general theory of relativity of 1915, which describes gravity as the bending of space and time by mass. In 1916, Karl Schwarzschild found the first exact solution of Einstein's equations that describes a black hole. The name "black hole" was popularised by the American physicist John Archibald Wheeler in 1967.

Anatomy of a Black Hole

At the centre of a black hole lies the singularity, a point where matter is thought to be crushed to infinite density and the known laws of physics break down. Around it is a boundary called the event horizon, which is the point of no return.

What Is an Event Horizon?

The event horizon is the boundary around a black hole beyond which nothing can escape, because the escape velocity there is equal to the speed of light. Anything that crosses the event horizon, whether matter, light or a spacecraft, is trapped forever and falls toward the singularity. The size of the event horizon is given by the Schwarzschild radius, which is proportional to the black hole's mass; for a black hole with the mass of the Sun, it would be about 3 kilometres, and for the Earth it would be about 9 millimetres. Outside the event horizon, a black hole acts like any other object with the same mass, and planets could orbit it safely at a distance. Around many black holes there is a bright, hot swirling disc of gas called an accretion disc, which glows strongly in X-rays before the matter falls in.

How Does a Black Hole Form?

Stellar-mass black holes form when a massive star dies. A star shines because nuclear fusion in its core produces outward pressure that balances the inward pull of gravity. When a star more than about 20 times the mass of the Sun runs out of fuel, fusion stops and gravity wins: the core collapses in a fraction of a second, and the outer layers are blown away in a huge explosion called a supernova. If the leftover core is heavier than about three times the mass of the Sun, no known force can stop its collapse, and it becomes a black hole. Smaller stars such as the Sun end as white dwarfs, while stars of intermediate mass leave neutron stars. The Indian-born astrophysicist Subrahmanyan Chandrasekhar calculated in 1930 that a white dwarf cannot exceed about 1.4 solar masses, a value known as the Chandrasekhar limit, and he received the Nobel Prize in Physics in 1983. Supermassive black holes, with millions or billions of times the Sun's mass, sit at the centres of most large galaxies, and scientists are still studying how they grew so large.

Types of Black Holes

Stellar-mass black holes have masses of a few to a few dozen times the Sun's mass. Intermediate-mass black holes, with hundreds to hundreds of thousands of solar masses, are rarer and harder to find. Supermassive black holes have millions to billions of solar masses. The Milky Way's central black hole, Sagittarius A*, has about 4 million solar masses and lies about 26,000 light-years from Earth. The galaxy Messier 87 hosts a black hole of about 6.5 billion solar masses. Primordial black holes are hypothetical small black holes that could have formed in the early universe.

How Do We Detect Black Holes?

Because black holes are invisible, scientists detect them by their effects. They observe stars orbiting an invisible object, X-rays from hot gas in accretion discs, and gravitational waves, ripples in space-time produced when two black holes merge. The first direct detection of gravitational waves was made by the LIGO observatories in the United States on 14 September 2015 from a merger of two black holes about 1.3 billion light-years away, announced in February 2016; the work earned the 2017 Nobel Prize in Physics for Rainer Weiss, Barry Barish and Kip Thorne. Cygnus X-1 was the first strong black hole candidate, found in the 1960s and 1970s. The 2020 Nobel Prize in Physics honoured Roger Penrose for showing black holes are a robust prediction of relativity, and Reinhard Genzel and Andrea Ghez for discovering the supermassive black hole at the centre of our galaxy.

First Image of a Black Hole

The first image of a black hole was released on 10 April 2019 by the Event Horizon Telescope (EHT) collaboration. It showed the supermassive black hole at the centre of galaxy Messier 87 as a bright ring of glowing gas surrounding a dark shadow. The EHT links radio telescopes around the world to act like one Earth-sized telescope. On 12 May 2022 the EHT released the first image of Sagittarius A*, the black hole at the centre of the Milky Way.

Hawking Radiation

In 1974, Stephen Hawking proposed that black holes slowly emit a faint radiation due to quantum effects, now called Hawking radiation. Over extremely long times this would cause black holes to shrink and evaporate. It has not been observed yet.

India and Black Hole Research

India contributes through Chandrasekhar's legacy, ISRO's AstroSat observatory, which has observed X-rays from black hole systems, and the LIGO-India project, planned in Hingoli, Maharashtra, which will join the global gravitational-wave network. Scientists at the Inter-University Centre for Astronomy and Astrophysics (IUCAA) in Pune and the Raman Research Institute in Bengaluru study black holes. Indian researchers were part of the LIGO discovery team.

Challenges and Future

Open questions include what happens at the singularity, the information paradox, and how supermassive black holes formed so early in the universe. Future telescopes and gravitational-wave detectors will study them in more detail.

Quick Facts

- The event horizon is the boundary around a black hole beyond which nothing can escape.
- Stellar black holes form when massive stars collapse after a supernova.
- The first image of a black hole (in galaxy M87) was released on 10 April 2019.
- The Milky Way's central black hole is Sagittarius A*, imaged in May 2022.
- Subrahmanyan Chandrasekhar won the 1983 Nobel Prize in Physics for work on stellar evolution.